

SMARTBAND V001

PCB Component & BOM Analysis

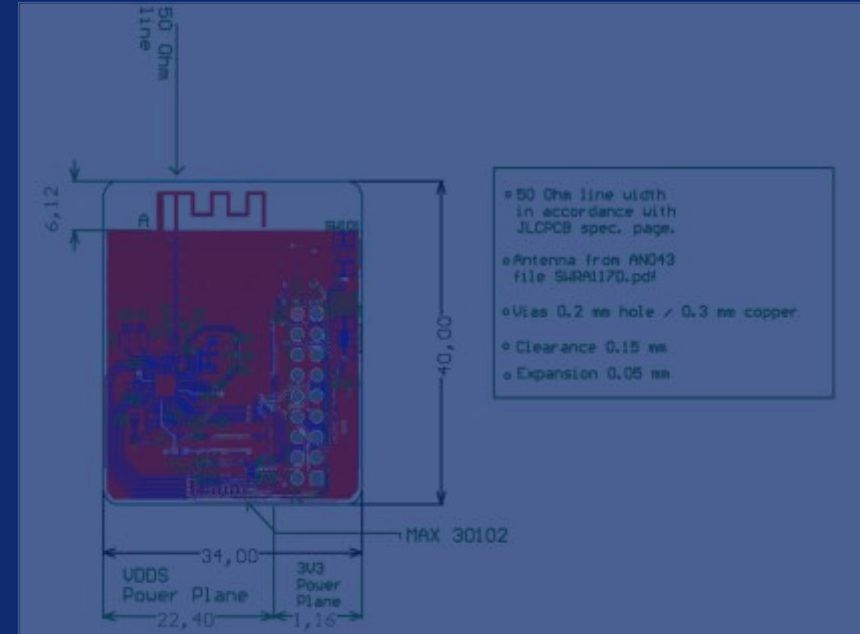
Band01_Rev1 — Hardware Procurement Documentation

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Task: Task 0 — PCB Review & Procurement



PROJECT OVERVIEW

Objective

Review the Band01_Rev1 PCB design, validate all components, update the BOM, communicate with providers, and place the manufacturing order.

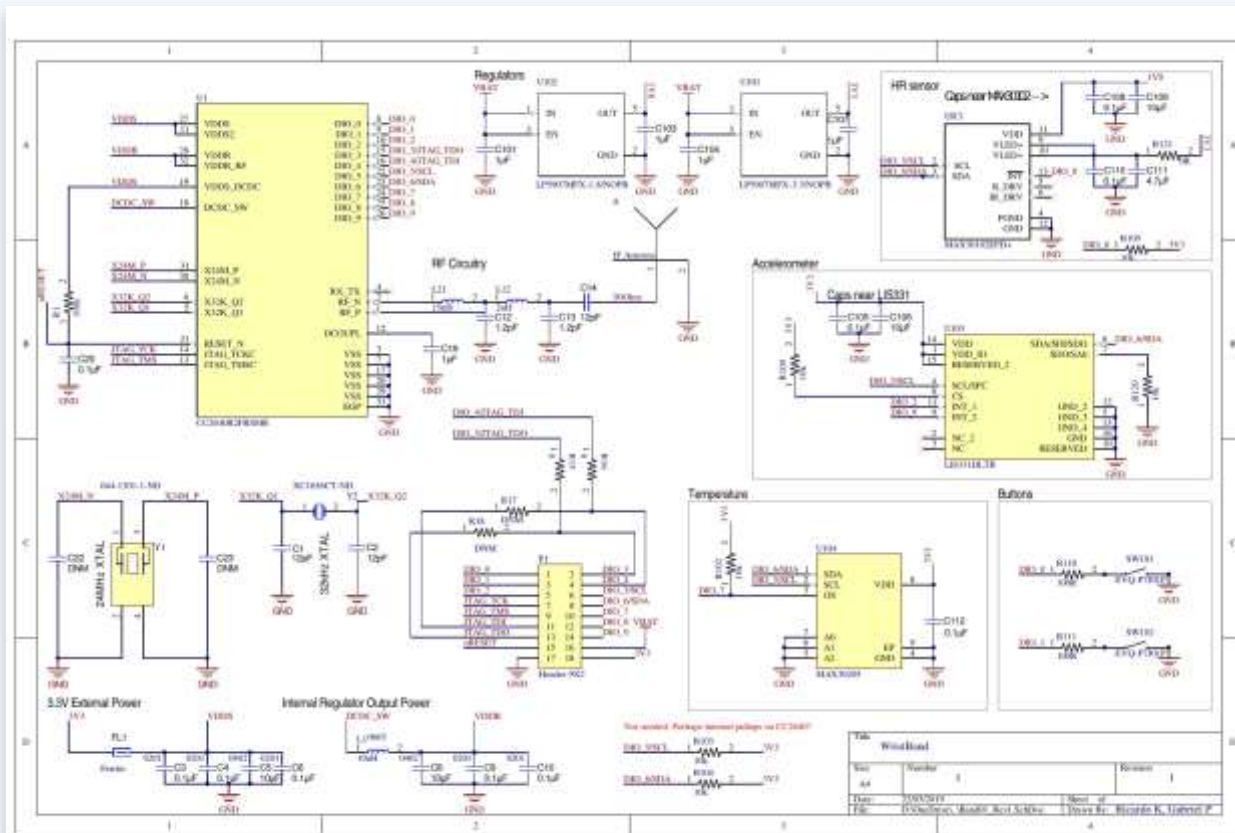
Platform

CC2640R2F BLE SoC
(ARM Cortex-M3, 48 MHz)
Bluetooth 5.0
Wrist-worn form factor
34 mm × 40 mm PCB

Research Context

Based on the MWHD framework by Pinheiro et al. (2022) — *Frontiers in Human Neuroscience*. Multi-sensor health data via BLE.

PCB SCHEMATIC — Band01_Rev1



U1 — CC2640R2F

BLE 5.0 SoC / MCU

UIC1 — MAX30102

HR & SpO2 Sensor

U103 — LIS331D

3-axis Accelerometer

U104 — MAX30205

Temperature Sensor

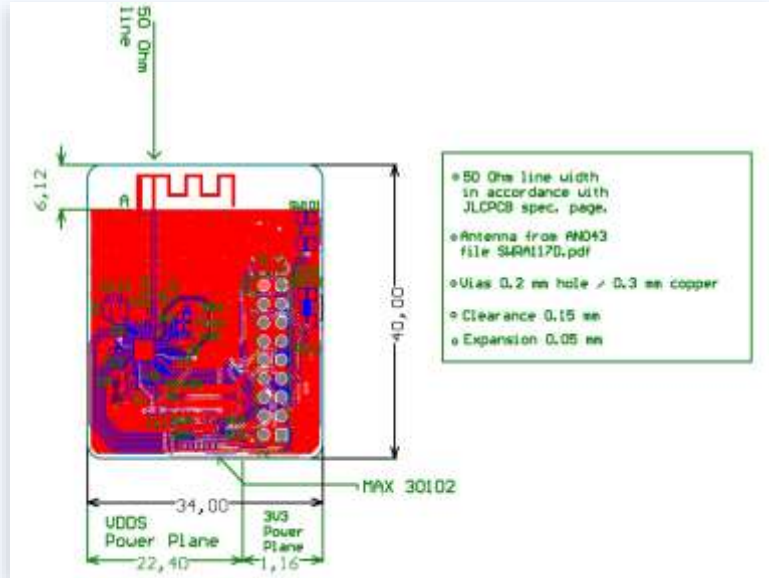
U101/U102 — LP5907

3.3V & 1.8V LDO Regs

Y1/Y2 — Crystals

24 MHz & 32.768 kHz

PCB LAYOUT & PHYSICAL SPECIFICATIONS



Dimensions

34 × 40 mm

Layers

2-layer PCB

Via Hole

0.2 mm / 0.3 mm Cu

Clearance

0.15 mm

Expansion

0.05 mm

RF Impedance

50 Ω

VDD5 Plane

22.40 mm

3V3 Plane

1.16 mm

Antenna

PCB trace (AN043)

KEY SENSOR SUBSYSTEMS

MAX30102EFD+

Heart Rate & SpO2

UIC1

18-bit ADC, integrated LED driver
Optical reflectance sensor
I2C interface

LIS331DLTR

Accelerometer

U103

3-axis MEMS, $\pm 24g$ range
Motion & activity detection
I2C interface

MAX30205MTA+T

Temperature

U104

$\pm 0.1^{\circ}\text{C}$ accuracy
Digital body temp sensor
I2C interface

CC2640R2FRSMR

BLE SoC / MCU

U1

ARM Cortex-M3 @ 48 MHz
Bluetooth 5.0, 128KB Flash
JTAG / SWD debug

BOM REVIEW & COMPONENT ANALYSIS

BOM Materials				
Item	Designator	Comment	Quantity	UOM
1	A	IF Antenna	1	IF
2	C2, C4, C5, C6, C7	1000000000	5	1000
3	C8, C9, C10, C11, C12	1000000000	5	1000
4	C13, C14, C15, C16	1000000000	4	1000
5	C17	1000000000	1	1000
6	C18, C19, C20, C21	1000000000	4	1000
7	C22, C23	1000000000	2	1000
8	C24, C25	1000000000	3	1000
9	C26	1000000000	1	1000
10	C27, C28	1000000000	2	1000
11	U1	1000000000	1	1000
12	U2	1000000000	1	1000
13	U3	1000000000	1	1000
14	U4	1000000000	1	1000
15	U5	1000000000	1	1000
16	U6	1000000000	1	1000
17	U7, U8, U9	1000000000	2	1000
18	U10, U11, U12, U13	1000000000	5	1000
19	U14	1000000000	1	1000
20	U15, U16	1000000000	2	1000
21	U17, U18	1000000000	2	1000
22	U19, U20	1000000000	2	1000
23	U21	1000000000	1	1000
24	U22, U23	1000000000	2	1000
25	U24	1000000000	1	1000
26	U25	1000000000	1	1000
27	U26	1000000000	1	1000
28	U27	1000000000	1	1000
29	U28	1000000000	1	1000
30	U29	1000000000	1	1000



33

Total Components

5

Items Needing Action

4

DNP / DNM Items

29

Line Items in BOM

Components Requiring Action:

IF Antenna (A)

PCB trace antenna used — marked DNP

P1 Header

Upgraded to Samtec HTSW-109-07-G-D

C22, C23

DNP optional bypass caps — retained

R17, R18

DNM — JTAG pull resistors not needed

Y1 Crystal

Sourced muRata equiv. for availability

Original BOM.pdf

PCBWay ASSEMBLY QUOTATION

Product No.: T-2Y3W994845A | 3 Assembled Sets

Component Cost

\$116.33

Assembly Cost

\$88.00

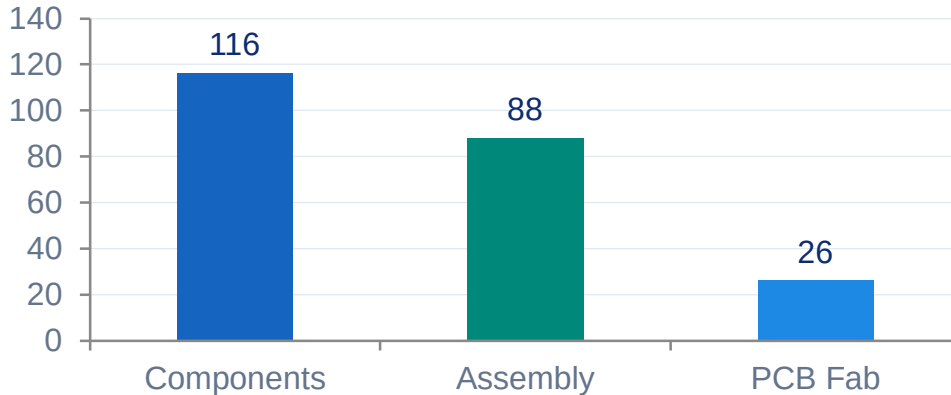
PCB Fabrication

\$25.97

TOTAL (3 sets)

\$230.30

Cost Breakdown (3 Sets)



Order Details

Unit Cost / Board:	≈ \$76.77
Lead Time:	26–28 working days
Quantity Ordered:	3 assembled sets
DNP Items:	A, C22/C23, R17/R18
Assembly Type:	SMT + Through-hole
Platform:	PCBWay.com

AISLER PLATFORM — FINAL ASSEMBLED PCB



The screenshot shows the Aisler BOM workspace for the 'Hochschule Herms-Lippstadt' project. It displays a list of components with columns for Name, Part, Comment, Footprint, Price, Impact, and Stock. The components are categorized into 'Needs assigning (5 components)' and 'Assigned (52 components)'. The 'Assigned' section shows components like '09H03 3C8021 54K15' and '09H03 3C8021 54K15' with their respective quantities and prices.

Name	Part	Comment	Footprint	Price	Impact	Stock
Needs assigning (5 components)						
A.	09H03 3C8021 54K15	IF_Antenna	IF			
C22, C23	09H03 3C8021 54K15	0201				
D10, D11	09H03 3C8021 54K15	0201				
Assigned (52 components)						
D1, D2	09H03 3C8021 54K15	0201		€10		
C3, C4, C6, C9, C10, C20	09H03 3C8021 54K15	0201		€10		

Aisler BOM — HSHL Workspace

Pricing Tiers (Aisler — Amazing Assembly):

Qty	Total Price	Unit Price
3	€376.69	€125.56
5	€439.65	€87.93
10	€590.66	€59.07
50	€1,893.43	€37.87
100	€3,249.84	€32.50
200	€6,183.19	€30.92
500	€14,134.49	€28.27

- ✓ 52 of 57 components assigned in Aisler (HSHL workspace)
- ✓ Prices include sourcing, handling, and assembly fees
- ✓ Offers valid for 24 hours from quotation

PROCUREMENT PROCESS

1

Received design ZIP from supervisor (GitHub repo)

2

Reviewed schematic — extracted all 33 components

3

Cross-referenced BOM vs. schematic — identified 5 issues

4

Researched availability on Mouser, DigiKey, LCSC, Aisler

5

Resolved component substitutions & part number updates

6

Prepared updated BOM — submitted to PCBWay for quotation

7

Received PCBWay quote: \$230.30 / 3 sets, 26–28 day lead time

8

Registered project on Aisler (HSHL workspace)

9

Assigned all available components in Aisler (52/57)

10

Order confirmed — PCB fabrication & assembly placed

CONCLUSION & NEXT STEPS

Task 0 — Deliverables Completed:

- ✓ Complete PCB schematic review (33 components)
- ✓ BOM validation & 5-item correction
- ✓ Provider communications (PCBWay + Aisler)
- ✓ Updated BOM submitted & quotation received
- ✓ Manufacturing order confirmed — 3 sets

Upcoming Steps:

- Await PCB delivery (26–28 day lead time)
- Inspect & test assembled boards on receipt
- Set up Code Composer Studio (CCS) for CC2640R2F
- Flash firmware — validate BLE communication
- Begin sensor integration & data acquisition
- Task 2: Firmware analysis & development